

OVERVIEW OF THREE FIELD METHODS FOR IMPROVING COVERAGE OF ADDRESS-BASED SAMPLES FOR IN-PERSON INTERVIEWS

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Address frames based largely on United States Postal Service lists can have shortcomings for in-person surveys in areas with non-locatable addresses. To give every housing unit a known, nonzero probability of selection, field methods may be implemented to improve frame coverage. Three coverage improvement methods currently in use in the survey research industry are compared—Enhanced Listing, Check for Housing Units Missed (CHUM), and Address Coverage Enhancement (ACE)—so that the reader may understand their similarities and differences. Such a comparison is useful for research practitioners and survey designers who need to consider nuances in frame construction methods that weigh cost, effectiveness, and ease of implementation.

KEYWORDS: ACE; CHUM; Coverage enhancement procedure; Dependent listing; Enhanced listing; Missed housing unit.

1. INTRODUCTION TO ABS FRAME SUPPLEMENTATION

Scientific surveys must be based on sampling frames that provide reasonable coverage of the target population to avoid the risk of bias. Consider the target population of households or persons who live in residential housing units in the United States, not including institutional housing or non-institutional group quarters. The ideal frame for such a population would be a list of all housing units. Historically, the primary method of obtaining a frame of housing units was to construct a multistage area probability design (Hansen and Hurwitz 1943; Kish 1965) where

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housing unit addresses were collected in the field through a process referred to as field enumeration (FE) or listing (See [Kish \[1965\]](#) for a description of the process. See [Harter, Eckman, English, and O'Muirheartaigh \[2010\]](#) for a recent example.) Due to the high cost of field enumeration, and with the aid of technological advancements, alternative approaches have been developed to prepare frames of housing units for sampling.

Following the 1990 Census, the US Census Bureau began a program of updating the file of housing units found during the census (the Address Control File), which led to the Master Address File (MAF) for the 2000 census ([United States Census Bureau 2000](#)). The MAF has been updated regularly ever since ([Kennel 2006](#)). Some of the Census Bureau's methods for updating the MAF became known as dependent listing (i.e., updating an existing list through field canvassing). The Census Bureau had for some time used a variety of methods to improve the coverage of their address lists, such as combining lists from multiple sources (e.g., prior lists with building permits) followed by in-field validation ([Liu 2008](#)). [Hansen, Hurwitz, and Madow \(1953\)](#) described field work to enumerate stores not already on an existing list for the Survey of Retail Stores in 1949. The Census Bureau's initiatives with pre-existing lists of sample units and addresses paved the way for the use of pre-existing address lists in depending listing methods by other organizations.

Today, address-based sampling (ABS) studies generally refer to surveys of residential populations whose frames are derived from vendor-provided address lists, usually from the small number of vendors who qualify to receive regular updates from the US Postal Service's Address Management System through a Computerized Delivery Sequence (CDS) license ([US Postal Service 2013](#)) or a Delivery Sequence File Second Generation (DSF²) license ([American Association for Public Opinion Research 2016](#)). The vendor file updated with active mail delivery sequence addresses, sometimes in combination with a supplemental Postal Service address list of mostly inactive addresses called the No-Stat file ([Shook-Sa, Currivan, McMichael, and Iannacchione 2013](#)), nominally includes all US mail delivery points, including essentially all occupied, vacant, and seasonal housing units in the United States ([Iannacchione 2011](#)). Although there is not always a one-to-one correspondence between delivery points and housing units, mailing addresses often serve as reasonable proxies for the physical locations of housing units. Research has shown that the number of mailing addresses on the address files from CDS-qualified vendors compares favorably with housing unit estimates from the US Census Bureau ([Iannacchione 2011](#)). Thus, for most surveys with mail contact, the coverage of the ABS frame is generally not a concern. For in-person surveys discussed here, the availability of address files allows for the central development of sampling frames without the need for costly field visits prior to sample selection. For this reason, ABS has been increasingly viewed as a lower-cost alternative to field enumeration for in-person surveys of the general population ([Iannacchione 2011](#)).

Nonetheless, most sampling frames have some coverage error. For example, traditional FE list frames contain neither the addresses of housing units built after the list frame was compiled nor units accidentally missed during enumeration. In addition, some addresses that do not exist or are not part of a selected area may be included erroneously. For ABS frames, certain types of postal delivery points (e.g., PO Boxes, Rural Route Boxes) are not suitable for in-person studies because the corresponding housing units cannot be physically located based on their mailing addresses. Addresses suitable for in-person fieldwork are commonly referred to as “city-style” addresses, and they consist of a unit’s number and street name, apartment number where applicable, city, state, and ZIP code. ABS frames for in-person surveys have been shown to suffer from undercoverage in rural areas with a higher proportion of PO Boxes, Rural Route Boxes, simplified addresses (just city, state, and ZIP code) and other unlocatable addresses (Dohrmann, Han, and Mohadjer 2007; Iannacchione, Morton, McMichael, Cunningham, Cajka et al. 2007; O’Muircheartaigh, English, and Eckman 2007; DiSogra, Dennis, and Fahimi 2010; Link 2010; Shook-Sa et al. 2013). Often such unlocatable addresses are simply removed from the frame due to the unknown association with housing units.

For clustered in-person surveys and other ABS surveys conducted in local geographical areas, the addresses are assigned latitude and longitude coordinates through a process called geocoding, and the geographical points are assigned to census blocks and area segments. Sometimes addresses are erroneously assigned to the wrong blocks and segments. If housing units assigned to sample segments but physically located outside the segments are excluded as ineligible, and if housing units that are physically in the sample segments are erroneously assigned elsewhere and not detected, coverage error is introduced. Coverage error is a greater problem in areas where the segments are small and geocoding is less accurate (Morton, Iannacchione, McMichael, Cajka, Curry et al. 2007; Eckman and English 2012).

Traditional FE frames had often been supplemented by a missed housing unit procedure during the interviewing stage of the study to ensure that any new construction or units erroneously omitted from the FE list frame had a chance of selection. The most common procedure was the Half-Open Interval (HOI) method (Stephan 1936; Deming 1960; Kish 1965), also known as the predecessor-successor method (Hansen, Hurwitz, and Jabine 1963; Singh 2004), in which the interviewer searched from a sampled housing unit up to but not including the next housing unit on the ordered frame list, and any new housing units found in that interval were also selected into the sample. By linking the previously missed housing unit to the one listed just before it on the ordered list, the probabilities of selection for missed housing units were set to be the same as the sampled unit. This method required a well-specified listing sequence or *path of travel* so that the interviewer could follow the same path as the field enumerator whose information was used to create the FE frame. One

issue with the HOI method for ABS is that the address lists are in mail delivery sequence order, which does not lend itself to well-defined half-open intervals because of the tendency of postal lists to cross streets and jump from block to block (McMichael, Ridenhour, and Shook-Sa 2008). In addition, while survey methodologists have long had reservations about the successful implementation of the HOI procedure (Kish and Hess 1958), results from a small-scale test confirmed that listers have trouble implementing the HOI to find addresses actually missing from frames, while often erroneously adding addresses not missing (Eckman and O'Muircheartaigh 2011).

In a response to the above issues, survey researchers have developed procedures to supplement vendor address files to obtain more complete address frames. A topic-contributed session at the November 2013 meeting of the Federal Committee on Survey Methodology (FCSM) highlighted the methods used by three major survey houses. English, Dekker, and O'Muircheartaigh (2013) discussed a method called Enhanced Listing (dependent listing) for improving the frame prior to sample selection. Dohrmann and Sigman (2013) described an area-linkage approach known as ACE, for "address coverage enhancement," which effectively subsamples area segments for field personnel to review for missed housing units. McMichael, Shook-Sa, and Ridenhour (2013) presented a method called Check for Housing Units Missed (CHUM), which captures missed housing units after the sample has been selected. This paper provides brief overviews of the three methods presented in that session and provides some guidance for selecting the appropriate method(s) for an in-person ABS study.

All of the above methods contain common elements. First, all require a person to visually inspect the housing units in the selected geographies. Such field visits are relatively costly but, at the time these methods were developed, were considered more timely and accurate than any methods conducted in a central office using internet resources (Dreiling, Trushenski, Kayongo-Male, and Specker 2009; Curtis and English 2013). (The US Census Bureau's plans for updating addresses for the 2020 Census [United States Census Bureau 2016] may indicate to the industry that it is time to take another look at central office methods.) Second, all methods require the field personnel to follow a well-defined path of travel. The specific paths of travel are interchangeable among methods, and any well-structured path of travel that is unambiguously clear, complete, and not redundant is suitable. None of the above methods uses the mail delivery sequence order for the path of travel, which is known to be discontinuous within selected geographies. Third, all methods require listed housing units to be geocoded to census blocks prior to any field visits.

We note that traditional FE is still used today (e.g., the National Survey on Drug Use and Health, Center for Behavioral Health Statistics and Quality 2016) and is an option for well-budgeted studies. At the other extreme, designers may opt for using the vendor address list without any additional supplementation in the field whatsoever. Alternatively, the vendor list may be

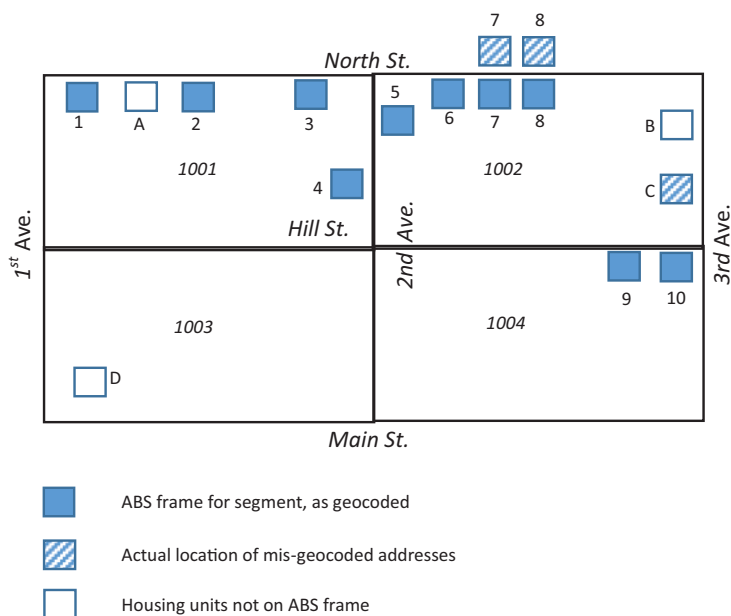


Figure 1. ABS Frame.

supplemented with the subset of active addresses in the No-Stat file. One study found that adding only certain types of addresses from the No-Stat file improved coverage of eligible housing units modestly without incurring much loss of efficiency (Shook-Sa et al. 2013). Furthermore, many national, in-person ABS studies now typically use a mixture of methods for improving coverage, tailoring the method to the expected coverage provided by the address lists for the selected geographies.

Section 2 provides a brief look at Enhanced Listing (English et al. 2013). Section 3 provides a summary of the CHUM procedure (McMichael et al. 2013). Section 4 gives a brief overview of the ACE procedure (Dohrmann and Sigman 2013). Section 5 compares and contrasts features of the methods for those considering which method might be appropriate for a particular study.

All three of the methods discussed are illustrated by the hypothetical segment in figure 1. In figure 1, shaded squares for housing units numbered one through ten are on the ABS frame and were geocoded into the boundaries of this segment. Units 7 and 8 are actually outside the designated blocks, and the striped squares show their actual locations. Unit C was incorrectly geocoded outside the segment. Units A, B, and D were not on the ABS list, even though all three are physically located within the segment blocks. These three units are shown in the segment blocks with open squares. That is, figure 1 shows the

housing units' actual physical locations, "ground truth" if you will, along with the ten units (1–10) that geocoded into the ABS frame for this segment.

2. ENHANCED LISTING

NORC at the University of Chicago developed Enhanced Listing as a method of preparing a complete frame of housing units prior to sample selection (O'Muircheartaigh et al. 2002; O'Muircheartaigh et al. 2007; O'Muircheartaigh, English, Lattermer, Eckman, and Dekker 2009). Enhanced Listing, also known as dependent listing, has been used by NORC for several studies, including the National Children's Study in two counties, the General Social Survey, the Survey of Consumer Finances, and Amerispeak (English, O'Muircheartaigh, Dekker, Bilgen, Fiorio et al. 2012). Enhanced Listing has also been used by the Census Bureau for the Current Population Survey and by the University of Michigan for the National Survey of Family Growth (Lepkowski, Mosher, Davis, Groves, and Van Hoewyk 2010; Eckman and Kreuter 2011).

Enhanced Listing is a lower-cost alternative to traditional field enumeration because the address list is not built entirely from scratch. In the field, the lister starts with a designated starting point, such as an intersection, and traverses all blocks in a sample segment in the prescribed path of travel, checking to see whether each housing unit encountered is on the ABS frame. If not, the housing unit is added to the frame. If an address on the list does not correspond to any housing unit on the ground in the segment, the address is removed from the frame. To be clear, each segment is defined by the boundaries of census blocks, and the enhanced frame is intended to be a complete list of all housing units and only those units within the segment's geographical boundaries.

Although early instances of Enhanced Listing were conducted with paper listings of ABS addresses for comparison, the process was greatly enhanced by mobile electronic devices that contained and mapped the ABS listings (Dekker, English, Winfrey, and Seeger 2013).

When the sample of housing units is selected, a housing unit's probability of selection is based on the segment probability, the number of housing units selected, and the number of housing units on the enhanced frame. Because the entire frame for selected segments is updated, the segments can be reused for subsequent surveys, provided the selected addresses are appropriately excluded.

Figure 2 illustrates the changes that are made to the ABS frame to complete the enhanced frame for this segment. Housing units seven and eight are removed, and housing units A–D are added, resulting in twelve housing units on the frame.

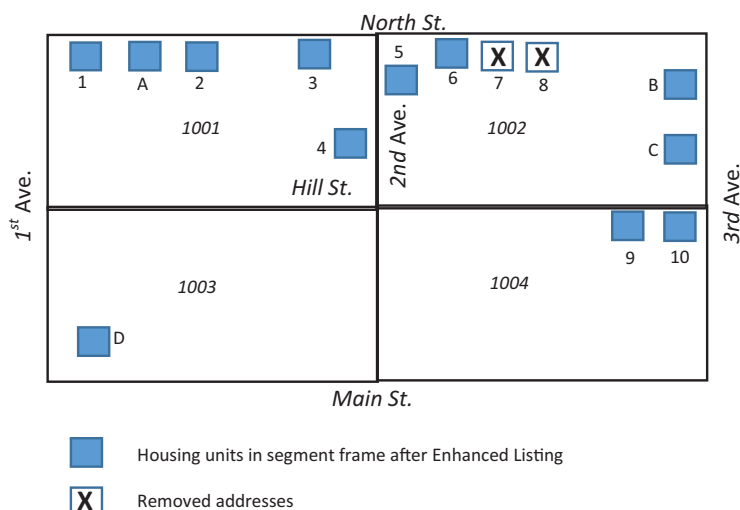


Figure 2. Enhanced Frame.

3. CHUM

The Check for Housing Units Missed (CHUM), developed at RTI International, has been used for the 2008 American National Election Study (Lupia, Krosnick, Luevano, DeBell, and Donakowski 2009), where it increased coverage by 5.1 percentage points. Most recently, the CHUM was used for the 2015 Residential Energy Consumption Survey sponsored by the US Energy Information Administration and a Tobacco Panel recruitment project for the FDA. The method has evolved somewhat since its original incarnation (McMichael et al. 2008), and now consists of two main components (McMichael et al. 2013; Shook-Sa, Harter, McMichael, Ridenhour, and Dever 2016). These components are implemented after sample selection and before, or along with, data collection.

The first component (CHUM1) is like the HOI procedure in that the area after a selected housing unit is searched in the prescribed path of travel. The search continues until either another housing unit on the ABS frame for that block is encountered or the entire block is searched. If another housing unit is encountered in the search that is not on the ABS frame for that block, it is added to the sample, and it has the same probability of selection as the originally sampled housing unit. If many are found clustered together, a subsample of the missed housing units can be added to the sample, and the probabilities adjusted accordingly.

CHUM1 differs from HOI in that the original ABS frame is not ordered according to the prescribed path of travel. As mentioned previously, the mail delivery sequence order is not suitable as the path of travel for supplementing

the frame, and it is not computationally simple to put the frame in path of travel order. The next housing unit found following the path of travel may be on the ABS frame, but not next in ABS frame order, complicating the task. Thus, it is useful to have the frame for the block in electronic format to make it easier to check if a found housing unit is on the frame.

CHUM1 is the only search method discussed in this paper that begins with a sample housing unit rather than a block feature such as a corner or intersection. If that housing unit was incorrectly geocoded into the sample segment, it is no longer in the sample, and the CHUM1 search beginning with that unit is not conducted.

CHUM1 will not discover missed housing units on blocks with no frame addresses, so the CHUM2 procedure was developed to search a sample of blocks in a selected segment. The sample could consist of all blocks in a selected segment, all blocks in the segment with no ABS addresses, or a subset of blocks expected to have undercoverage. The selected blocks are given a starting point, and each block is searched as in CHUM1 until either an ABS housing unit is encountered in the block or the entire block is searched. If an ABS address is encountered, it already had a chance of selection, and any missed housing units on the block had a chance of being found in the CHUM1 procedure. If no ABS housing units are found, any housing units that are found in the block are added to the sample, possibly with subsampling. The probability of selection for the added housing units is the probability of the segment times the conditional probability of selecting the block for CHUM2, adjusted for any subsampling of found housing units.

Like the Enhanced Listing procedure, the CHUM procedure assumes that a segment consists of all housing units that truly exist within the boundaries of the blocks that comprise the segment. A housing unit found to be outside the segment boundaries is discarded as a frame error, even if it was selected for the sample. The probabilities of selection, however, are based on the frame at the time of sampling, geocoding errors and all. The entire segment is not searched, only intervals beginning with selected housing units (CHUM1) or designated starting points (CHUM2). If the ABS coverage is extremely low, then the amount of searching approaches that of Enhanced Listing; otherwise, the CHUM should be less burdensome to implement in the field.

Figure 3 illustrates the adjustments to the sample based on the CHUM procedure. Suppose units one and five were selected into the sample, and the third block was selected for CHUM2. Unit A is added to the sample through CHUM1, and unit D is added through CHUM2. The goal of CHUM is to give every housing unit a chance of selection, not to complete the frame. Therefore, units B and C are not added to the sample even though B was omitted from the ABS frame, and C was geocoded to a different segment. The frame had the original ten ABS units at the time of sampling, and the sample was augmented by CHUM after initial sample selection.

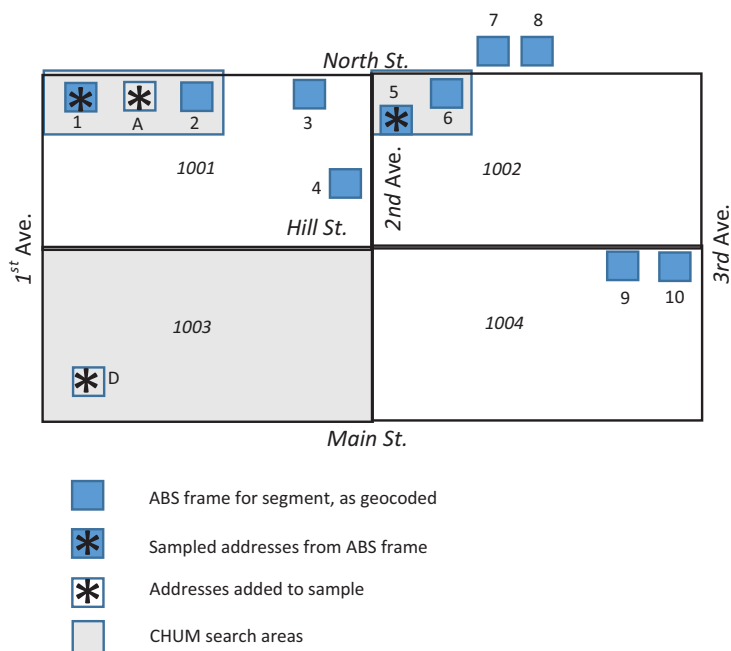


Figure 3. CHUM Sample.

4. ACE

Address Coverage Enhancement (ACE) was developed by Westat (Dohrmann and Sigman 2013; Kalton, Kali, and Sigman 2014), and the procedure also goes by the name CEP for “coverage enhancement procedure.” The ACE procedure has been used for the National Epidemiologic Survey on Alcohol and Related Conditions (NESARC-III) (Kalton et al. 2014). Theoretically, ACE can be performed before or after housing unit selection, but in practice has been done after selection of housing units and just before screening and interviewing. Like Enhanced Listing, entire area segments defined by census geography are searched; however, not all sample segments are searched. In one version, ACE is performed in an area segment if and only if the first housing unit in the segment in frame order is selected into the sample.

Unlike Enhanced Listing and CHUM, the sampling frame is defined to consist of all housing units that geocode into the segment boundaries, regardless of where they actually are located, along with any housing units found within the area segment boundaries that are not already on the ABS frame in any other segment. ACE makes the distinction between the “list segment,” comprised of addresses from the list frame that geocoded into the “area segment,” defined by the geographic boundaries. Thus, ACE requires the entire ABS frame in

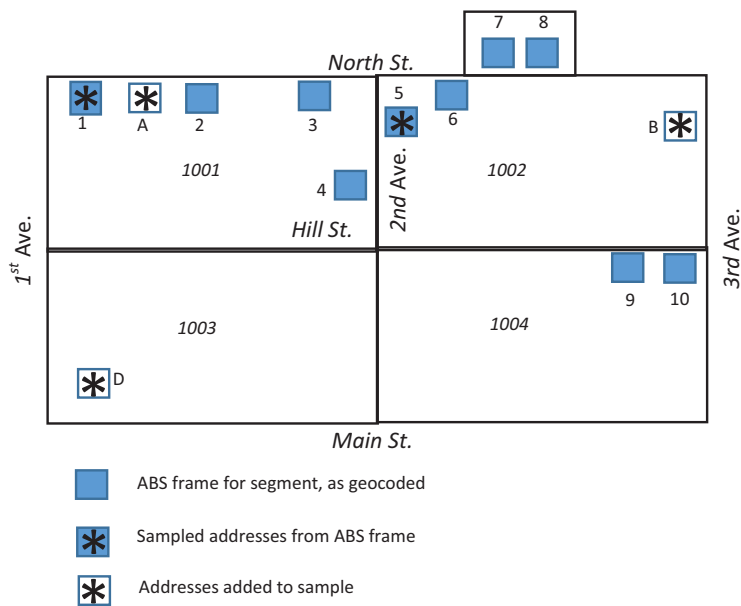


Figure 4. ACE Frame.

and surrounding the selected segments to be available in electronic format to facilitate checking. Conversely, geocoding errors for addresses on the ABS frame do not have to be corrected, thus saving field work. The corrected frame for an ACE segment consists of the ABS units that geocode into a segment, as well as the non-ABS units that are found within the segment's nominal physical boundaries. The corrected sample consists of the original sample units and the units added to the segment (assuming the number is small enough that no subsampling is required).

The ACE procedure is not performed in all sample segments. Unlike CHUM, the missed units are not linked to the prior unit in the path-of-travel order, but all missed units in a segment may be linked to a single unit in the list frame (e.g., the first listed unit). Then the ACE procedure is performed in the segment if and only if that first listed unit is selected into the sample (whether or not the first listed unit is physically within the area segment boundaries). All the units found by the procedure and added to the sample will have the same probability of selection as the first unit; [Kalton et al. \(2014\)](#) discuss ways of adjusting the subsampling probabilities for the segments and the subsampling rate of found housing units such that the workload and the weights are controlled. The ACE procedure requires searching the entire segment, not selected intervals as in CHUM.

[Figure 4](#) illustrates the revisions to the segment frame and sample based on ACE. Suppose units one and five were selected into the sample. No housing

units are removed from the frame, and housing units A, B, and D are added to the frame and the sample. Housing unit C is not added because it is on the ABS frame, but was geocoded into a neighboring segment. The resulting frame has thirteen housing units.

5. COMPARE AND CONTRAST

Enhanced (dependent) Listing has been used in one form or another for more than two decades (O'Muircheartaigh, Eckman, and Weiss 2002; United States Census Bureau 2006). CHUM was developed a few years later (McMichael et al. 2008). The ACE procedure is a relative newcomer (Dohrmann et al. 2012), borrowing features from the other methods and putting a unique twist on the concepts. All three methods have been used successfully for large national surveys. All three methods involve some field work, the magnitude of which is not fully known in advance. The methods have different strengths and weaknesses, as shown in [table 1](#). Users should be aware that doing nothing about coverage error may be the best choice in some situations; certainly, doing nothing incurs no additional expense. Also, traditional field enumeration is still in use for some studies.

Features of one method could be adapted for another method. For example, all methods could subsample the segments for which coverage adjustment procedures are followed, with corresponding impacts on costs and weight variability. Both CHUM and ACE could be done in the same trip as screening and interviewing, although both procedures work better if the field personnel do coverage adjustment procedures before screening begins; otherwise, the personnel's attention to coverage adjustment is shared with screening and interviewing, which may be given higher priority.

Because Enhanced Listing is completed prior to housing unit selection, sample sizes are not affected. CHUM and ACE, however, might cause sample sizes within a geographical area to vary, and subsampling of found housing units may cause weights to vary. Initial sample housing units may be selected with the same probability in all geographical areas, but areas that are then supplemented will naturally have higher sample counts. To reduce workloads in the supplemented geographical areas, the initial sample counts in these areas may be reduced, based on expected coverage rates, and the expected number of missed housing units may be found, on average; however, there is no guarantee that the expected number of missed housing units for each geographical area will be found accordingly. Either way, the supplemented geographical areas may have uneven workloads for field interviewers. If large clusters of housing units are found, they can be subsampled to control sample sizes, but subsampling increases the weights of the found housing units. The trick is finding an acceptable balance of sample sizes and weight variability through sample selection probabilities, subsampling of geographical areas, and

Table 1. Comparing and Contrasting ABS Coverage Supplementation Methods

Constraints and considerations	Coverage supplementation method		
	Enhanced listing	CHUM	ACE
Cost comparison?	Less than traditional FE, but still searches entire segments	Less than enhanced listing, likely comparable with ACE	Less than enhanced listing, likely comparable with CHUM
Control of sample size?	Sample size not affected	Less control over sample size with added units	Less control over sample size with added units
Size of segments?	Size a direct influence on cost	Size less relevant because not searching the entire segment	Size a direct influence on cost, unless number of ACE segments compensates
Definition of segments?	Based on census geographies and HUs that actually are located within them	Based on census geographies and HUs that actually are located within them	Area segments based on census geographies, and list segments based on HUs that geocode within the area segments, geocoding error notwithstanding
Concepts for geocoding error?	Geocoding error corrected in selected segments	Geocoding error corrected in search intervals	Geocoding error expected and part of design
Applied to which segments?	Usually limited to segments that do not have extremely high coverage, based on coverage threshold	Usually limited to segments that do not have extremely high or extremely low coverage, based on coverage thresholds	Usually limited to segments that do not have extremely low coverage, based on coverage thresholds
Subsampling of segments?	Possible	Possible	Only segments that meet the rule are enhanced
Design includes very low coverage areas?	Enhanced listing still applies	May revert to traditional FE below a coverage threshold	May revert to traditional FE below a coverage threshold

Continued

Table 1. Continued

Constraints and considerations	Coverage supplementation method		
	Enhanced listing	CHUM	ACE
Blocks with no frame addresses?	Listed if part of sample segment	Chance of selection through CHUM2 if part of sample segment	Searched if part of ACE segment
Timing?	As late as possible before housing unit selection	After housing unit selection, either at the time of data collection or just before	After housing unit selection, either at the time of data collection or just before
Focus of field personnel?	Completely separate trip, so primary focus	Could be in a separate trip or in the data collection trip; more focused and less risk of error if scheduled before interviewing starts	Could be in a separate trip or in the data collection trip; more focused and less risk of error if scheduled before interviewing starts
Technology or equipment required?	Computerized (preferred) or paper	Computerized (preferred) or paper	Computerized (preferred) or paper

subsampling of found housing units. [Kalton et al. \(2014\)](#) describe some ways of manipulating these factors for ACE, and the ideas could be adapted for CHUM.

Many details in implementation, such as extent of training, segment sizes, field tools (e.g., tablets, electronic maps and forms), and the quality review process affect costs. But generally, the expected ordering from most expensive to least expensive is traditional FE, Enhanced Listing, ACE and CHUM, and using the address list as is. The relative costs of ACE versus CHUM depend in part on the adjustments to linkage and subsampling rates in ACE—whether the amount of searching in whole ACE segments is more or less than searching intervals in all CHUM segments. Because ACE is not performed in all sample segments, fewer personnel need to be trained, or the cadre of trained personnel might travel less, so searching time is not the only cost driver. But cost is not the only consideration, either. Study requirements for coverage, timeliness, ease of implementation, control over sample size, and other factors enter into the decision. [Table 1](#) highlights some of the differences so that the methods can be evaluated in a particular context.

It is common for ABS studies to divide the selected area segments into groups based on expected coverage rates and apply different frame-building and supplementation methods for different coverage levels (American Association for Public Opinion Research 2016). For example, the segments with high coverage may not need any frame supplementation. Segments with extremely low coverage might use traditional field enumeration or enhanced listing. In between, any one of the supplementation methods described herein might be used. Decisions about estimating coverage and determining the coverage thresholds for different treatments are beyond the scope of this paper.

Where any coverage enhancement procedure is used, adequate training is key to success as any of the above methods will be ineffective if not implemented correctly.

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